

AI Engineering Intensive

From LLM Foundations to Production-Grade AI Systems

COMPLETE MERGED SYLLABUS · WEEKS 1–12 · 84 DAYS

6 July 2026 Series start	8 November 2026 Series end	84 days · 12 weeks 18 calendar weeks
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Phase	Weeks	Days	Dates	Status
Track 0	—	—	Self-study (Python, Pandas, NumPy, OOP, SQL)	Reference
Phase 1	1–4	1–28	6 Jul – 2 Aug 2026	■ COMPLETE
■■ Pause	—	—	3 Aug – 13 Sep 2026 (42 days)	In progress
Phase 2	5–12	29–84	14 Sep – 8 Nov 2026	Upcoming

Topics. Python & SQL Foundations · Machine Learning · Deep Learning & CNNs · Computer Vision · NLP · LLMs · Tokenization · Embeddings · Transformers · Quantization · Fine-Tuning · Serving · RAG · Hybrid Retrieval · Agents · Tool Use · Multi-Agent · LangChain · LangGraph · CrewAI · AG2/AutoGen/Agno · MCP · Evals · RAGAS · TruLens · LangSmith · Guardrails · Harness Engineering · Loop Engineering · Context Engineering · FastAPI · System Design · Multimodal · Security & Red Teaming · LLMOps · Cloud (AWS/Azure/GCP) · Docker · CI/CD · No-Code AI Platforms

MERGE LOCK m1 (2 Aug 2026). An incoming 4-month (16-week) syllabus has been absorbed into these 12 weeks — no weeks added. A term-by-term audit against the 28-PDF Phase 1 corpus found ~60% already delivered in Days 1–28. Genuine additions: classical ML, deep learning & computer vision, classical NLP, FastAPI depth, named agentic frameworks, cloud infrastructure detail, and the complete no-code block. Python/Pandas/NumPy/OOP and SQL are preserved in full as Track 0 (self-study).

Merge Evidence — measured, not assumed

Incoming block	Cov	Tot	%	Verdict
M2W3 Transformers & LLMs	7	8	88%	ALREADY TAUGHT
M1W2 SQL	6	7	86%	ALREADY TAUGHT

Incoming block	Cov	Tot	%	Verdict
M3W3 RAG + Agentic Frameworks	12	16	75%	ALREADY TAUGHT
M4W3 Cloud Infrastructure	9	12	75%	ALREADY TAUGHT
M3W4 Production AI & Evaluation	8	11	73%	ALREADY TAUGHT
M3W2 LLM & Prompt Engineering	10	14	71%	ALREADY TAUGHT
M3W1 Software Engineering / FastAPI	7	12	58%	PARTIAL
M2W3 Classical NLP	6	12	50%	PARTIAL
M1W3 ML Fundamentals	7	16	44%	PARTIAL
M2W1 Deep Learning / CNN	5	13	38%	PARTIAL
M1W1 Python / Pandas / NumPy / OOP	2	8	25%	NEW
M4W4 No-Code & AI Productivity	2	14	14%	NEW

TRACK 0 · FOUNDATIONS (Self-Study — Not Teaching Days)

Preserved verbatim from the incoming syllabus · upstream of Day 1

Foundations Week 1 · Python Setup, Core Programming & Pandas Basics

Topics. Python Installation & Environment Setup · Explore IDEs: VS Code, Jupyter, Colab · Jupyter Notebook Basics · Python Basics: Syntax, Variables, Strings, Input/Output · Loops, Functions, Conditional Statements · Python Libraries Exploration: Pandas, NumPy, Seaborn, Plotly · Python Data Structures: Lists, Tuples, Dictionaries, Sets · DataFrames, Series, Indexing, Selection · Data Input/Output · Data Cleaning Techniques · Handling Missing Data · Data Visualization with Matplotlib, Seaborn, Plotly · NumPy Arrays, Broadcasting, Indexing · OOPS Basics, Classes, Objects, Constructors, Data Input/Output · Inheritance, Polymorphism, Encapsulation, Abstraction

Free Resources. 01 Python Tutorial For Beginners (13 Hours) (*AI Coach John's Material*) · GenAI Support: ChatGPT

Foundations Week 2 · Structured Query Language

Topics. Intro to SQL, Basic SQL syntax, data types, primary keys, and constraints · Creating and Inserting into DB · DQL and DML Commands and filtering data · Aggregate Functions and Subqueries · Group By and Joins, types of joins · Window Functions and Stored Procedures

Free Resources. 01 SQL Full Course (4.5 Hours) (*AI Coach John's Material*) · GenAI Support: ChatGPT

Already applied: SQL is exercised continuously in the Days 13–27 Text-to-SQL capstone — joins, aggregates, subqueries, window functions and CTEs all appear in generated and validated queries.

PHASE 1 — COMPLETE ■

Weeks 1–4 · Days 1–28 · 6 July – 2 August 2026

Verified: 28 PDFs · 549 pages · 371 Eagle Eye terms · 0 gaps · 0 FAANG-rule violations.

WEEK 1 · FOUNDATIONS — Days 1–7 · 6–12 Jul · Model Selection Dashboard (MINI)

Day	Date	Topic — the one thing that matters
1	Mon Jul 6	Tokenization — tokens are the billing unit <i>and</i> the context unit
2	Tue Jul 7	Embeddings — similarity is geometry; cosine over normalized vectors
3	Wed Jul 8	Transformer Architecture — attention is quadratic; why long context costs
4	Thu Jul 9	Training Lifecycle — pre-train vs fine-tune vs prompt: different problems
5	Fri Jul 10	Scaling Laws & Prompt Engineering — CoT is the cheapest accuracy
6	Sat Jul 11	Structured Output — constrain the format, don't ask for it
7	Sun Jul 12	Model Selection Dashboard — weighted criteria, not just "which is best"

WEEK 2 · EFFICIENCY — Days 8–14 · 13–19 Jul · Domain Fine-Tune + Quantize (PORTFOLIO)

Day	Date	Topic — the one thing that matters
8	Mon Jul 13	Quantization — 4×–8× memory reduction, minimal quality loss
9	Tue Jul 14	LoRA & PEFT — adapters make specialization economically viable
10	Wed Jul 15	Full Fine-Tuning — catastrophic forgetting is real and measurable
11	Thu Jul 16	Model Serving — continuous batching: 210 → 900 q/hr
12	Fri Jul 17	Evaluation — pick the metric that matches the failure mode
13	Sat Jul 18	Efficiency — PagedAttention enables prefix sharing
14	Sun Jul 19	Text-to-SQL Foundation — execution accuracy, not exact match

■ **The running capstone begins Day 13** — a production Text-to-SQL system built continuously through Day 27.

WEEK 3 · RETRIEVAL & AGENTS — Days 15–21 · 20–26 Jul · End-to-End Text-to-SQL (CAPSTONE)

Day	Date	Topic — the one thing that matters
15	Mon Jul 20	RAG Foundations — retrieval adds knowledge; nothing else does
16	Tue Jul 21	RAG Indexing Pipelines — fingerprints make re-indexing idempotent
17	Wed Jul 22	RAG Retrieval Deep Dive — dense + BM25 + RRF + MMR + rerank
18	Thu Jul 23	RAG Generation — CoT + faithfulness + self-correction + citations
19	Fri Jul 24	Agents and Tool Use — agent = loop; MAX_STEPS is a safety control
20	Sat Jul 25	Multi-Agent Systems — specialize by role → mix model sizes → cut cost
21	Sun Jul 26	Week 3 Capstone — ablation justifies architecture: RAG = +53.5pp

WEEK 4 · PRODUCTION — Days 22–28 · 27 Jul – 2 Aug · Production Hardening (PORTFOLIO)

Day	Date	Topic — the one thing that matters
22	Mon Jul 27	Production LLM Operations — your failure mode is a 200 OK with a wrong answer
23	Tue Jul 28	Prompt & Model Versioning — a prompt is a model weight you can edit
24	Wed Jul 29	Cost Optimization — 73% of input tokens are identical every request
25	Thu Jul 30	Security & Compliance — assume injection succeeds
26	Fri Jul 31	Multi-Modal Systems — an image <i>is</i> tokens; extract once, index forever
27	Sat Aug 1	Fine-Tuning for Production — doesn't add knowledge; buys cost and latency
28	Sun Aug 2	Week 4 Capstone & Close — the handoff <i>is</i> the deliverable

Incoming Topics Already Delivered in Phase 1

Incoming topic	Delivered on
Tokenization	Day 1

Incoming topic	Delivered on
Word Embeddings; Cosine Similarity, Semantic Search & Text Similarity	Day 2
Attention Mechanism, Self-Attention & Encoder-Decoder; Transformer Architecture	Day 3
Large Language Models (LLMs) & Foundation Models	Days 3–4
LLM Fundamentals & Inference Pipeline	Days 4, 11
Prompt Engineering Principles & Patterns; Zero/One/Few-Shot; Chain-of-Thought	Day 5
Structured Outputs; Output Parsers & Schema Validation	Day 6
Decoding Parameters (Temperature, Top-P, Max Tokens)	Days 5–6
Function Calling & Tool Calling	Days 6, 19
Retrieval-Augmented Generation; Document Loaders & Chunking Strategies	Days 15–16
Embedding Models & Vector Databases	Days 15–16
Semantic Search & Reranking	Day 17
Agent Memory & Planning Strategies	Days 19–20
Prompt Injection & Prompt Security	Days 18, 25
AI Safety & Guardrails	Days 22, 25
Building Production-Ready LLM Applications	Days 14, 21, 22
Docker for AI Applications (<i>introduced</i>)	Day 11
Context Window & Token Management (<i>cost framing</i>)	Day 24

■ ■ PAUSE

3 August – 13 September 2026 · 42 days · no teaching days

On resume, before Day 29: read the Resume Brief (Day 28 §5) · re-read Days 19–20 · run one canary evaluation.
First three actions: (1) buy accuracy headroom — two prompt fixes worth ~3pp; (2) deploy Tier 1 security — `llm_query_role` + RLS, 20 min; (3) wire the two CI gates.

PHASE 2

Weeks 5–12 · Days 29–84 · 14 September – 8 November 2026

WEEK 5 · AGENTS, HARNESS & LOOP ENGINEERING + AGENTIC FRAMEWORKS

Days 29–35 · 14–20 Sep 2026 · Project: AI Research Agent with Tool Calling (PORTFOLIO)

Day	Date	Topic
29	Mon Sep 14	Agent Architectures Beyond ReAct
30	Tue Sep 15	Harness Engineering I — Tool Design
31	Wed Sep 16	Harness Engineering II — Responses, Errors & Environment
32	Thu Sep 17	Loop Engineering I — Termination & Budgets
33	Fri Sep 18	Loop Engineering II — Recovery, Sub-Agents & Human-in-the-Loop Design
34	Sat Sep 19	Agentic Frameworks I — LangChain Fundamentals & LangGraph Workflows
35	Sun Sep 20	Week 5 Project — AI Research Agent with Tool Calling

Resources carried in: Loop Engineering (■■■■■) (AI Coach John) · Complete LangGraph Tutorial in Tamil (Free & Open Source) (AI with Akash) · LangChain for Beginners: Your First AI Agent Explained in Tamil (AI Coach John)

WEEK 6 · CONTEXT ENGINEERING, MCP & MULTI-AGENT FRAMEWORKS

Days 36–42 · 21–27 Sep 2026 · Project: Multi-Agent Business Analyst System (PORTFOLIO)

Day	Date	Topic
36	Mon Sep 21	Context Engineering I — The Context Budget & Token Management
37	Tue Sep 22	Context Engineering II — Compaction & Progressive Disclosure
38	Wed Sep 23	MCP (Model Context Protocol) Fundamentals
39	Thu Sep 24	Building a Production MCP Server
40	Fri Sep 25	Agentic Frameworks II — CrewAI, AG2 / AutoGen / Agno & Multi-Agent Collaboration
41	Sat Sep 26	Agent Memory Architectures & Planning Strategies
42	Sun Sep 27	Week 6 Project — Multi-Agent Business Analyst System

Resources carried in: MCP Crash Course (2 Hours) (AI Coach John) · Build AI Agents with Crew AI Framework (AI with Thiru) · Autogen Agentic AI Framework (Aai Tech)

WEEK 7 · PRODUCTION AI SYSTEMS, EVALUATION & OBSERVABILITY

Days 43–49 · 28 Sep – 4 Oct 2026 · Project: Eval & Observability Platform (PORTFOLIO)

Day	Date	Topic
43	Mon Sep 28	AI System Architecture & Orchestration
44	Tue Sep 29	LLM-as-a-Judge & Eval Harness Design
45	Wed Sep 30	Hallucination Detection & Evaluation
46	Thu Oct 1	RAG Evaluation — RAGAS & TruLens
47	Fri Oct 2	LangSmith for Debugging & Observability
48	Sat Oct 3	AI Safety & Guardrails in Production
49	Sun Oct 4	Week 7 Project — Eval & Observability Platform

WEEK 8 · AI APPLICATION ENGINEERING & AGENTIC SYSTEM DESIGN

Days 50–56 · 5–11 Oct 2026 · Project: End-to-End AI Backend + Scalable Agent Runtime (PORTFOLIO)

Day	Date	Topic
50	Mon Oct 5	AI Application Architecture · REST APIs, HTTP Methods & API Integration · JSON, JSON Schema & Structured Data
51	Tue Oct 6	FastAPI Fundamentals & API Development · Request Handling, Response Models & Error Handling
52	Wed Oct 7	Authentication, API Keys & Environment Variables · File Uploads, Streaming Responses & Async APIs
53	Thu Oct 8	Database Integration (SQLite/PostgreSQL Basics) · AI API Integration (OpenAI, Gemini, Claude)
54	Fri Oct 9	Agentic System Design — Runtime, State Management & Queueing
55	Sat Oct 10	Scaling, Backpressure, Failure Isolation & Concurrency
56	Sun Oct 11	Week 8 Project — End-to-End AI Backend Development using FastAPI + Scalable Agent Runtime

Resources carried in: API — Step by Step — Build, Secure and Deploy | Full Practical Guide (Applied with AI) · Complete FastAPI Tutorial in Tamil — Build your own RestAPI (AI with Akash)

WEEK 9 · MACHINE LEARNING & DEEP LEARNING FOUNDATIONS

Days 57–63 · 12–18 Oct 2026 · Project: ML & DL Project Development & Deployment (PORTFOLIO)

Day	Date	Topic
57	Mon Oct 12	ML Fundamentals — Types of ML (Supervised, Unsupervised & Reinforcement Learning) · Features, Labels, Training/Testing Data & Train/Test Split · Classification, Regression & Clustering Algorithms
58	Tue Oct 13	Model Validation using Cross-Validation · Evaluation Metrics: Accuracy, Precision, Recall, F1-Score & ROC-AUC · Confusion Matrix & Handling Class Imbalance
59	Wed Oct 14	Overfitting, Underfitting & Regularization Techniques · Feature Engineering & Data Preprocessing · Feature Scaling, Normalization & Data Preparation · End-to-End ML Workflow using Scikit-learn
60	Thu Oct 15	Introduction to Deep Learning, Neural Networks & Perceptron · Activation Functions, Loss Functions & Optimizers · Forward Propagation, Backward Propagation & Gradient Descent
61	Fri Oct 16	Epochs, Batch Size, Learning Rate & Hyperparameter Tuning · Batch Normalization, Dropout & Vanishing Gradient Problem
62	Sat Oct 17	CNNs & Image Classification · Pretrained Models, Transfer Learning & Fine-Tuning · SOTA Models & Modern DL Applications · End-to-End Model Development using TensorFlow/Keras or PyTorch
63	Sun Oct 18	Week 9 Project — ML & DL Project Development & Deployment

Week 9 Project Set — preserved verbatim

Project Title	Category	Industry / Application
Predictive Maintenance for Industrial Equipment	Regression	Manufacturing / Industry 4.0
Credit Risk Assessment & Loan Default Prediction	Classification	Banking & FinTech
Customer Segmentation using K-Means & DBSCAN	Clustering	Retail / E-Commerce / Marketing
Brain Tumor Detection & Segmentation using U-Net + CNN	Image Classification / Segmentation	Healthcare / Medical AI

Project Title	Category	Industry / Application
Video Anomaly Detection for Smart Surveillance	Video Classification / Anomaly Detection	Smart Cities / Security

Resources carried in: Machine Learning (■■■■■■■■) in 25 Minutes: Absolute Beginner Crash Course (*AI Coach John's Material*) · Industry-Standard AI, Data Science & Machine Learning Full Course in Tamil for Beginners (*Hire Ready*) · AI/ML (■■■■■■■■): Complete Deep Learning RoadMap Explained in 15 Mins (*AI Coach John's Material*) · Deep Learning Series in Tamil (*Adi Explains*)

WEEK 10 · NLP, MULTIMODAL, REASONING & RED TEAMING

Days 64–70 · 19–25 Oct 2026 · Project: NLP & Multimodal Projects + Red Team Assessment (PORTFOLIO)

Day	Date	Topic
64	Mon Oct 19	Introduction to Natural Language Processing (NLP) & Text Preprocessing · Tokenization, Stop Words, Stemming & Lemmatization
65	Tue Oct 20	Bag of Words (BoW), Count Vectorizer & TF-IDF · N-Grams, POS Tagging & Named Entity Recognition (NER)
66	Wed Oct 21	Word Embeddings: Word2Vec, GloVe & Vector Embeddings · End-to-End NLP Pipeline using Hugging Face Transformers
67	Thu Oct 22	Vision-Language Models, OCR Pipelines & Document Understanding at Scale
68	Fri Oct 23	Chart & Diagram Reasoning · Reasoning Models & Test-Time Compute
69	Sat Oct 24	Red Teaming, OWASP LLM Top 10 & Responsible AI Governance
70	Sun Oct 25	Week 10 Project — NLP & Multimodal Projects + Red Team Assessment

Week 10 Project Set — preserved verbatim

Project Title	Category	Industry / Application
AI Resume Screening & Candidate Matching System	NLP + Semantic Search	HRTech / Recruitment
Legal Document Question Answering using RAG	LLM + RAG	LegalTech / Enterprise AI
Intelligent Customer Support Chatbot with Vector Search	Conversational AI + LLM	Customer Support / SaaS

Resources carried in: What is NLP? Beginners to Advanced RoadMap 2025 in Tamil (AI Coach John's Material) · NLP Playlist (Chill and Grow)

WEEK 11 · CLOUD DEPLOYMENT, AI INFRASTRUCTURE & LLMOps

Days 71–77 · 26 Oct – 1 Nov 2026 · Project: Deploy a Production-Ready AI Service (PORTFOLIO)

Day	Date	Topic
71	Mon Oct 26	Introduction to AWS, Azure & Google Cloud Platform · Compute Services (EC2, Azure VM, Compute Engine) · Storage Services (S3, Blob Storage, Cloud Storage)
72	Tue Oct 27	Docker Fundamentals & Containerization · Docker for AI Applications
73	Wed Oct 28	Deploying FastAPI Applications · Deploying AI Applications on Cloud · Serverless AI Deployments
74	Thu Oct 29	CI/CD Basics for AI Applications · Monitoring & Logging · Cost Optimization & Cloud Best Practices
75	Fri Oct 30	LLMOps — Model Registries, Automated Eval Gates & Inference Optimisation (TensorRT-LLM, ONNX)
76	Sat Oct 31	Synthetic Data Generation (Self-Instruct, Evol-Instruct) · Feature Stores & Data Lineage · Cost Engineering & Chargeback
77	Sun Nov 1	Week 11 Project — Deploy a Production-Ready AI Service

Resources carried in: AWS in Half Hour | Learn EC2 Basics in Tamil (Tamil Coding / AWS Tamil) · What is Docker? Explained in Tamil | Full Tutorial + Demo (Arivi by HCL GUVI) · AWS EC2 vs Azure Virtual Machine (■■■■■■■■■) (K. Mohamed Faizal) · Docker & Containerization in Tamil (Arivi by HCL GUVI) · AWS for Beginners in Tamil Playlist (Tamil Coding) · Azure Fundamentals in Tamil (K. Mohamed Faizal) · Google Cloud Platform (GCP) in Tamil (Tamil

Cloud / Community Resources) · CI/CD Pipeline in Tamil (*DevOps Tamil*) · FastAPI Deployment using Docker & Cloud (*English — Recommended if Tamil is unavailable*) · Deploying AI Applications on AWS using FastAPI & Docker (*English — Advanced*) · Dockers for Beginners (*Simply Byte*)

WEEK 12 · NO-CODE AI PLATFORMS & ENTERPRISE CAPSTONE

Days 78–84 · 2–8 Nov 2026 · Project: Enterprise Multi-Document RAG Assistant (CAPSTONE)

Day	Date	Topic
78	Mon Nov 2	Introduction to No-Code & Low-Code AI Platforms · Claude Projects & Claude Artifacts · ChatGPT Projects, GPTs & Canvas
79	Tue Nov 3	Microsoft Copilot & Microsoft 365 Copilot · Microsoft Copilot Studio & AI Agents (Basics) · Google Gemini Workspace & Gems
80	Wed Nov 4	AI Workflow Automation using n8n (Introduction) · AI Coding Assistants (GitHub Copilot, Cursor & Windsurf) · Rapid AI Prototyping & Business Use Cases · Choosing Between No-Code, Low-Code & Custom AI Development
81	Thu Nov 5	Capstone I — Architecture, ADRs & C4 Diagrams
82	Fri Nov 6	Capstone II — Build & Integration
83	Sat Nov 7	Capstone III — Evaluation, Security & Deployment
84	Sun Nov 8	Capstone IV — Enterprise AI Assistant Final Defence

Capstone Project — preserved verbatim

Project Title	Category	Industry / Application	YouTube Search
(Project 1) Enterprise Multi-Document RAG Assistant	RAG + LangChain	Enterprise Knowledge Management	Enterprise RAG LangChain LangGraph Project

#	Required Capstone Deliverable
1	Architecture Decision Records (ADRs) for every major choice (vector DB, model, framework, cloud)
2	C4 System Diagrams — Context, Container, Component, Code
3	Non-Functional Requirements — latency SLAs, availability, cost/request, scaling limits
4	Risk Assessment — failure modes, guardrail-bypass scenarios, residual risk
5	Deployment Plan — blue-green strategy, rollback, monitoring dashboards
6	Security Report — red-team results, guardrail coverage, compliance mapping

Resources carried in: Claude Projects & Claude Artifacts (■■■■■■■■■) (AI Coach John) · Google Gemini Workspace & Gems (■■■■■■■■■) (AI Coach John) · Microsoft Copilot & Microsoft 365 Copilot (■■■■■■■■■) (K. Mohamed Faizal) · Microsoft Copilot Studio & AI Agents (Basics) (English — Microsoft Official) · AI Workflow Automation using n8n (■■■■■■■■■) · AI Workflow Automation using n8n (■■■■■■■■■) — Part 2 · GitHub Copilot (■■■■■■■■■) (Tamil Tech / Tamil Coding Community)

Placement Index — Every Incoming Block

The incoming 4-month syllabus mapped to its destination. Nothing dropped; no weeks added.

Incoming	Title	Placed at
M1W1	Python Setup, Core Programming & Pandas Basics	Track 0 (self-study)
M1W2	Structured Query Language	Track 0; applied Days 13–27
M1W3	Machine Learning Fundamentals & Core Concepts	Days 57–59
M1W4	ML Project Development & Deployment	Day 63
M2W1	Deep Learning Fundamentals & Neural Networks	Days 60–62
M2W2	DL Project Development & Deployment (Image/Video)	Day 63
M2W3	NLP & Large Language Models	Days 64–66 + Days 1–4
M2W4	NLP Projects & Deployment	Day 70
M3W1	Software Engineering Foundations for AI Applications	Days 50–53
M3W2	LLM Engineering & Prompt Engineering	Days 4–6, 18, 25 + Day 36
M3W3	RAG Systems & Agentic AI Frameworks	Days 15–20 + Days 33, 34, 40, 41
M3W4	Production AI Systems, Evaluation & Deployment	Days 43–48, 72, 73
M4W1	Enterprise AI Project Development — 1	Week 12 Capstone
M4W2	Enterprise AI Project Development — 2	Days 35, 42
M4W3	Cloud Deployment & AI Infrastructure	Days 71–74
M4W4	No-Code AI Platforms & AI Productivity Tools	Days 78–80

What Was Compressed — stated plainly

Depth in security, multimodal and system design is reduced to create room for ML/DL/NLP foundations and the no-code block. The compressed material is not removed — it layers on days already delivered in Phase 1.

Original Phase 2 plan	Now	Justification
W8 Agentic System Design — 7 days	3 days (D54–56)	Day 20 covered orchestration, pools, parallel phases
W9 Multimodal — 7 days	2 days (D67–68)	Day 26 covered VLM architecture, pipelines, tables, staged eval
W10 Security — 7 days	2 days (D69–70)	Day 25 covered the full Tier 1–4 control hierarchy
W12 Capstone — 7 days	4 days (D81–84)	3 days redirected to the no-code block

Project Ladder — 13 Named Projects

Week	Project	Type
01	Model Selection Dashboard	MINI ■
02	Domain Fine-Tune + Quantize	PORTFOLIO ■
03	End-to-End Text-to-SQL	CAPSTONE ■
04	Production Hardening	PORTFOLIO ■

Week	Project	Type
05	AI Research Agent with Tool Calling	PORTFOLIO
06	Multi-Agent Business Analyst System	PORTFOLIO
07	Eval & Observability Platform	PORTFOLIO
08	End-to-End AI Backend + Scalable Agent Runtime	PORTFOLIO
09	ML & DL Project Development (5 projects)	PORTFOLIO
10	NLP & Multimodal + Red Team Assessment (3 projects)	PORTFOLIO
11	Deploy a Production-Ready AI Service	PORTFOLIO
12	Enterprise Multi-Document RAG Assistant	CAPSTONE

Daily Protocol

Every teaching day runs all four steps:

1. TEACH

the day's topic fully in chat - concept-first, plain English, FAANG examples, Architect's Lens, hands-on exercises
2. EAGLE EYE

two-pass verification: automated keyword scan, then MANDATORY snip verification on EVERY flag. Never declare a gap without snip verification; never declare coverage without confirmation. Standard: "Don't hallucinate and don't miss anything."
3. PDF

-> ~/Desktop/AI-ML/WeekN_DayNN_TopicName.pdf
4. PUSH

-> week-NN/notes/ on the GitHub repository

FAANG RULE (LOCKED)
All real-world examples use ONLY Meta, Apple, Amazon, Netflix, Google, Microsoft. Hard constraint on all content.

GenAI Support: ChatGPT (as noted throughout the incoming syllabus)

RESUME TRIGGER
Say "Day 29 Week 5 Start" on or after 14 September 2026.

Authoritative Sources

Document	Authority over
SCHEDULE.md	All dates. If any document disagrees, this file wins.
CURRICULUM_MASTER.md	All content, Weeks 1–12. Merged syllabus, Track 0, placement index.
CURRICULUM_PHASE2.md	Superseded — retained for the harness/loop/context audit rationale.
CURRICULUM_EXTENDED.md	Weeks 10 & 11 legacy detail.
Day 28 PDF §5	RESUME BRIEF — read first on 14 September.

Progress: 28 of 84 days complete (33%). Phase 1 sealed and verified. Phase 2 resumes Monday 14 September 2026 with Day 29.